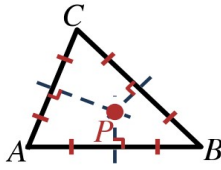
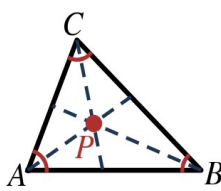
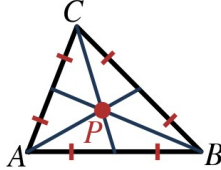
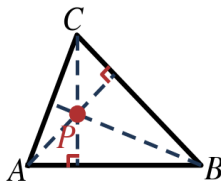


Points of Concurrency — A Reference Guide

Four centers, four constructions. This page tells them apart; the next two show where they land and how to choose.

Name	Made from	Where it lands	What it gives you	Picture
Circumcenter	the three PERPENDICULAR BISECTORS of the sides	inside · on · outside	equidistant from the three VERTICES — center of the circle through all three	
Incenter	the three ANGLE BISECTORS	always inside	equidistant from the three SIDES — center of the largest circle that fits inside	
Centroid	the three MEDIANS (vertex to opposite midpoint)	always inside	the balance point; it cuts every median in a 2 : 1 ratio from the vertex	
Orthocenter	the three ALTITUDES (vertex ⊥ opposite side)	inside · on · outside	no distance property — it is simply where the three altitudes meet	

How to Tell Them Apart in One Question

- Ask what the problem is measuring FROM the center. Distance to the three CORNERS → circumcenter. Distance to the three SIDES → incenter.
- Ask what was drawn. Lines that cut SIDES in half at a right angle → circumcenter. Rays that cut ANGLES in half → incenter. Segments from a vertex to the opposite MIDPOINT → centroid. Segments from a vertex PERPENDICULAR to the opposite side → orthocenter.
- Watch the trap: a MEDIAN goes to the midpoint, an ALTITUDE goes in perpendicular. They are the same segment only when the triangle is isosceles and you draw from the vertex angle.
- Only the centroid has a ratio to memorize: 2 : 1 from the vertex. Vertex-to-centroid is twice centroid-to-midpoint, and the centroid sits 2/3 of the way down each median.
- Only the circumcenter and the incenter have an equidistance theorem. The orthocenter has none — do not invent one.

Where Each Center Lands

The centroid and the incenter are always inside. These two are not — and the picture explains why.

	Acute triangle	Right triangle	Obtuse triangle
Circumcenter			
	inside	ON the triangle — the midpoint of the hypotenuse	outside
Orthocenter			
	inside	AT the right-angle vertex	outside

Why the Right-Triangle Case Is Worth Memorizing

- In a right triangle the two legs are themselves altitudes, so the orthocenter is forced to the right-angle vertex — no drawing required.
- In a right triangle the circumcenter is the MIDPOINT OF THE HYPOTENUSE. That means the midpoint of the hypotenuse is the same distance from all three vertices, so the hypotenuse is a diameter of the circle through the triangle.
- Whenever a coordinate problem gives you a vertical side and a horizontal side, you have a right triangle — find the two midpoints, write $y =$ (a horizontal line) and $x =$ (a vertical line), and the circumcenter falls out in one step.

Triangle Inequalities — The Short Version

- Bigger side, bigger angle. The longest side of a triangle is opposite the largest angle, and the shortest side is opposite the smallest angle. To order one list, order the other.
- Triangle Inequality Theorem: the sum of any two sides must be GREATER than the third. Checking the two smallest is enough.
- Range for a missing third side: (difference of the two known sides) $< x <$ (sum of the two known sides).
- Indirect proof, in three moves: (1) assume temporarily that what you want to prove is FALSE, (2) reason until you hit something impossible, (3) conclude that the assumption failed, so the original statement is true.

Name: _____

Naming Practice

For each description, name the point of concurrency: CIRCUMCENTER, INCENTER, CENTROID, or ORTHOCENTER. Some ask for a short reason as well.

1. The point where the three perpendicular bisectors of the sides meet.
2. The point where the three angle bisectors meet.
3. The point where the three medians meet.
4. The point where the three altitudes meet.
5. Equidistant from the three vertices of the triangle.
6. Equidistant from the three sides of the triangle.
7. The center of the largest circle that fits inside the triangle.
8. The center of the circle that passes through all three vertices.
9. The balance point of a triangular sheet of metal.
10. Always inside the triangle, no matter what kind of triangle it is. (Name BOTH.)
11. Lands at the vertex of the right angle when the triangle is right.
12. Lands at the midpoint of the hypotenuse when the triangle is right.
13. Divides each of three segments in a 2 : 1 ratio from the vertex.
14. A cell tower must be the same distance from three towns.
15. A sprinkler must reach three straight fences equally.